

Classifier Choice Outweighs Representation Choice for Noise-Robust Twitter Sentiment Classification

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Abstract—Social media text is routinely corrupted by typing errors and dropped words, yet classical text classifiers are usually reported only on clean benchmark splits. This study measures that cost. Bag-of-Words (BoW) and TF-IDF representations were paired with Naive Bayes, a linear Support Vector Machine (SVM), and Multinomial Logistic Regression (MLR), trained on 3,999 clean tweets from the Twitter Entity Sentiment Analysis dataset and tested on 998 held-out tweets, clean and corrupted, at three noise severities with five trials each. MLR gave the best clean accuracy (46.9%, $\kappa = 0.275$) and lost only 2.4 percentage points under moderate noise. Classifier choice mattered far more than representation: for the SVM and MLR the two representations never differed by more than 1.4 points and that gap reversed sign under cross-validation, whereas Naive Bayes trailed the discriminative models by 10.3 to 18.1 points and fell to near-chance agreement ($\kappa = 0.046$) on noisy TF-IDF. Naive Bayes was also the only classifier the two representations separated, always in favour of BoW. The practical implication is that for a pipeline of this kind the weighting scheme is a second-order decision, while the choice of classifier family is the first-order one.

Index Terms—Text Classification, Bag-of-Words, TF-IDF, Naive Bayes, Support Vector Machine, Noise Robustness, Sentiment Analysis

I. INTRODUCTION

Sentiment classification of social media text is a well-established application of Natural Language Processing (NLP), and classical pipelines built on Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) representations remain widely used because they are computationally inexpensive, interpretable, and easy to deploy [1]. A recent benchmark across twenty datasets confirmed that TF-IDF

features with a linear classifier still match or beat far heavier models on many text-classification tasks [2]. Text collected from platforms such as Twitter, however, rarely resembles the clean, well-formed sentences on which these representations were developed. Users abbreviate words, omit letters, and misspell terms under the pressure of fast, informal typing, and such surface noise erodes the vocabulary overlap between training and test data that BoW and TF-IDF rely on to represent meaning.

Despite the popularity of these representations, classical machine-learning research rarely measures how much accuracy is lost when a trained model meets corrupted input, as opposed to reporting accuracy on one clean split. It is therefore unclear which classifier, paired with which representation, degrades least under noise. This study quantifies how BoW and TF-IDF, combined with Naive Bayes, a linear SVM, and Multinomial Logistic Regression (MLR), respond to controlled textual noise in a four-class Twitter sentiment task, and tests whether the resulting ordering survives repeated trials and cross-validation. The main contributions are as follows.

- A reproducible, multi-severity noise-injection protocol, applied only to held-out test text so that no model is trained on corrupted data, evaluated over six representation-classifier pairs with five independent trials per severity.
- An evaluation that reports Cohen’s Kappa and the majority-class baseline alongside accuracy, together with per-fold cross-validation, which separates differences that are stable under resampling from those that are not.

- Evidence that classifier choice dominates representation choice under noise, and the identification of TF-IDF with Naive Bayes as the one configuration that collapses to near-chance agreement once the input is corrupted.

The remainder of this article is organised as follows. Section II reviews related work, Section III describes the methodology, Section IV presents and discusses the results, and Section V concludes.

II. RELATED WORK

Supervised sentiment classification was placed on a firm empirical footing by Pang, Lee, and Vaithyanathan [3], who showed that classifiers trained on simple lexical features outperform hand-crafted rules. The approach reached social media soon afterwards: Neethu and Rajasree [4] built a Twitter-specific feature vector for product opinions, Gautam and Yadav [5] reported that adding WordNet semantics improved on a purely statistical representation, and the survey of Kharde and Sonawane [6] identified Naive Bayes, Maximum Entropy, and SVM as the dominant classical baselines. Qi and Shabrina [7] found that machine-learning models generalise across topics better than fixed sentiment lexicons, which supports the purely machine-learning pipeline adopted here.

More recent work compares the same three model families used in this study. Srikanth et al. [8] evaluated Bernoulli Naive Bayes, Linear SVC, and Logistic Regression on TF-IDF features and singled out Logistic Regression as the strongest of the three. Al Sarayrah et al. [9] added a BiLSTM baseline and found a classical RBF-kernel SVM slightly ahead of it (88.64% against 87.72%), with multinomial and complement Naive Bayes trailing that SVM by 8.4 and 7.2 points. Zhang [10] reported 95.12% for a Naive Bayes classifier once TF-IDF was augmented with domain n-grams and BERT-guided feature selection, against 84.43% for an SVM plus TF-IDF baseline, which suggests that the weakness of Naive Bayes is tied to the feature space it is given rather than being intrinsic to the model family. The benchmark of Reusens et al. [2] likewise confirms that simple TF-IDF pipelines remain competitive with embedding-based models. All of these comparisons use clean text.

Robustness to corrupted input has been studied mainly on the neural side. Belinkov and Bisk [11] showed that small amounts of synthetic and natural character-level noise sharply degrade neural machine translation, Moradi and Samwald [12] reported the same sensitivity in neural language models, and Aliakbarzadeh et al. [13] recently reached that conclusion for large multilingual models on naturally noisy text. Whether classical, non-neural classifiers behave the same way, and which representation-classifier pair holds up best, is the question addressed here. Table I summarises the studies discussed above and states how each differs from the present work.

III. METHODOLOGY

The experimental workflow of this study is shown in Fig. 1. It comprises six phases: i) dataset selection and stratified sampling, ii) text preprocessing, iii) BoW and TF-IDF feature

extraction, iv) the train-test split and classifier training, v) noise injection on the held-out test set, and vi) performance evaluation. Noise is injected only after the split, so that no model is ever trained on corrupted text and no test information reaches the training stage.

A. Selection of Data Collection Method

A quantitative, experimental design was adopted because the research question concerns the measurable behaviour of trained classifiers under controlled conditions. A public, pre-labelled corpus was used rather than live scraping, since the required sentiment labels already exist, no new annotator bias is introduced, and other researchers can reproduce the noise experiment on the same freely available data.

B. Dataset and Preprocessing

The Twitter Entity Sentiment Analysis dataset [14] was selected because it provides four sentiment classes across 32 entities and more than 70,000 usable tweets, which is large enough for stratified sampling while remaining computationally tractable. Rows with missing text and exact duplicates were removed from the 74,682 raw tweets, leaving 71,656, from which a stratified sample of 4,998 was drawn with a fixed random seed. Each tweet was then processed with the `tm` package in R [15] through lowercasing, punctuation and number removal, English stopword removal, Porter stemming, and whitespace normalisation. Table II summarises the result.

C. Feature Representation

Two document-term matrices were built from the preprocessed corpus: a raw-frequency BoW matrix, and a TF-IDF matrix following the classical weighting of Salton and Buckley [1],

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log\left(\frac{N}{df(t)}\right) \quad (1)$$

where $\text{tf}(t, d)$ is the frequency of term t in document d , N is the number of documents, and $df(t)$ is the number of documents containing t . Terms with sparsity above 0.99 were removed, yielding a vocabulary of 177 terms for both representations.

D. Noise Injection

A parameterised noise function was applied to the test text only, after the split. It performs random word deletion with probability p_{del} and adjacent-character transposition with probability $p_{swap} = 0.10$ on words longer than three characters, transferring the character-level scrambling used by Belinkov and Bisk [11] for neural machine translation to a classical pipeline. Three severities were tested, $p_{del} \in \{0.10, 0.20, 0.30\}$, each repeated five times with independent random draws; a fixed intermediate setting of $p_{del} = 0.15$ was used for the main clean-versus-noisy comparison. Corrupted text passed through the same preprocessing pipeline and was mapped onto the fixed training vocabulary.

TABLE I
REPRESENTATIVE RELATED WORK AND POSITIONING OF THIS STUDY

Work	Technique	Key Finding	Difference from This Work
Neethu & Rajasree [4]	Twitter feature vector + SVM	Machine-learning classification feasible for product-domain tweets	Single classifier; no representation or noise comparison
Gautam & Yadav [5]	BoW/TF-IDF + WordNet semantics	Semantic enrichment improved accuracy	Clean text only; no robustness test
Kharde & Sonawane [6]	Survey of lexicon and ML methods	NB, MaxEnt and SVM are the dominant baselines	Survey; no original noise experiment
Srikanth et al. [8]	BernoulliNB, LinearSVC, LR on TF-IDF	Logistic Regression strongest of the three	Clean text only; no repeated trials
Al Sarayrah et al. [9]	NB, SVM, BiLSTM on TF-IDF	RBF SVM 88.64% vs. BiLSTM 87.72%; NB 7–8 points below the RBF SVM	Product reviews; robustness not examined
Zhang [10]	TF-IDF + NB, BERT-guided features	NB reached 95.12% with enriched features; SVM + TF-IDF baseline 84.43%	News domain, not social-media text
Reusens et al. [2]	11 classifiers, 20 datasets	TF-IDF pipelines competitive with embeddings	Clean data throughout
Belinkov & Bisk [11]	Character noise in neural MT	Small noise levels sharply degrade translation	Neural MT, not classical classification

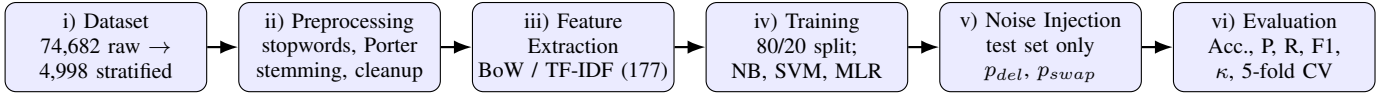


Fig. 1. The six phases of the experimental pipeline. Noise is injected only after the train-test split, so no model is ever trained on corrupted text.

TABLE II
DATASET SUMMARY

Property	Value
Raw tweets collected	74,682
After removing missing/duplicates	71,656
Stratified sample used	4,998
Class counts (Irrel./Neg./Neut./Pos.)	874 / 1,514 / 1,235 / 1,375
Vocabulary size (after filtering)	177 terms
Train / Test split	3,999 / 998
Majority-class baseline (Negative)	30.26%

E. Classification Algorithms

Three classical classifiers were trained on the clean training split for each representation: Naive Bayes [16], which estimates the class posterior under a conditional-independence assumption; a linear-kernel Support Vector Machine [17], which fits a maximum-margin hyperplane and is extended to four classes by one-against-one voting; and Multinomial Logistic Regression [18], which models the class posterior with a softmax link. Naive Bayes and the SVM were fitted with the `e1071` package and MLR with the `nnet` package [19], all with default hyperparameters, since the focus is the representation and noise comparison rather than hyperparameter optimisation. Because the term counts were supplied as numeric predictors, `e1071` estimates each class-conditional density as a Gaussian rather than as the multinomial event model normally preferred for count features [16]; the consequences of that choice are taken up in Section V.

F. Training Configuration and Evaluation Metrics

An 80/20 stratified split was created with `caret::createDataPartition` [20] and a fixed

seed, producing 3,999 training and 998 test tweets, and the same split was reused for every representation-classifier pair. Accuracy is reported together with precision, recall, and F1, computed per class in a one-vs-rest manner and macro-averaged as recommended for multi-class problems by Sokolova and Lapalme [21],

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad (2)$$

$$F_1 = \frac{2 \times P \times R}{P + R}$$

where TP , FP , and FN are read off the confusion matrix for the class under consideration. Because the classes are unbalanced, raw accuracy can be misleading, so Cohen’s Kappa [22] is also reported,

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (3)$$

where p_o is the observed agreement and p_e the agreement expected by chance from the class marginals, so that $\kappa = 0$ means performance no better than chance. All metrics were computed with `caret::confusionMatrix` [20], which also reports the majority-class No-Information Rate used as the baseline throughout, together with a one-sided exact binomial test of accuracy against that rate; the p -values quoted in Section IV come from that test. Finally, 5-fold cross-validation was run on the clean data to test whether the observed differences are stable or an artefact of one partition.

G. Ethical Considerations

The study uses only publicly available, pre-anonymised tweet text; no new personal data was collected and no re-identification was attempted. The main ethical concern is

validation reverses it in favour of TF-IDF (47.52% against 46.12%, Table VI). Naive Bayes is the exception, and the only one: BoW leads TF-IDF for it by 4.0 points on clean data, 6.2 under noise, and 4.2 on the cross-validated mean. That does not disturb the ranking, since Naive Bayes finishes behind both discriminative models either way. Comparing the slopes in Fig. 3 between $p_{del} = 0.10$ and 0.30, TF-IDF even degrades slightly more slowly than BoW for both discriminative classifiers (−2.2 and −2.0 points, against −2.9 and −2.8 for BoW), the opposite of what the sharpening effect of IDF weighting on rare terms might suggest. A gap of 1.4 points that flips sign under resampling is not a sound basis for choosing a representation.

The classifier gap behaves very differently. Naive Bayes trails MLR by 10.3 points on clean BoW, 11.8 points on noisy BoW, and 18.1 points on noisy TF-IDF, and it finishes last in all ten fold-by-representation cells of Table VI, behind whichever of the two discriminative models is nearer by between 7.5 and 16.0 points. Both models that beat it fit a decision boundary directly, whereas Naive Bayes models each feature independently given the class; the size and consistency of the gap suggest that this discriminative-versus-generative distinction, not the weighting scheme, governs performance in this feature space. This asymmetry is the central result of the study.

Fig. 4 draws the same six pairs as bars. Between the two panels only the Naive Bayes group changes appreciably; the SVM and MLR groups are near-copies of themselves. Within either panel the Naive Bayes pair sits in a band of its own, with a clear gap before the next bar. The single bar that drops beneath the baseline is Naive Bayes on noisy TF-IDF, where a weak classifier and the sharper weighting scheme reinforce each other.

D. Where the Errors Concentrate

Table IV gives the confusion matrix for the strongest configuration, MLR on BoW features on the clean test set; its 468 diagonal entries out of 998 reproduce the 46.9% accuracy of Table III. The residual errors are not spread evenly. The model issues only 109 Irrelevant predictions for 174 true Irrelevant tweets, a recall of 0.276, dispersing the remainder mainly into Positive (53 cases) and Negative (47 cases), while Positive is over-predicted with 354 predictions for 275 true instances. Applying (2) class by class gives Table V: Negative is handled best ($F1 = 0.528$), consistent with it being both the largest class and the one with the most distinctive vocabulary, and because macro-averaging weights all classes equally the weak Irrelevant class pulls macro F1 (44.9%) below accuracy (46.9%). Under noise every class loses ground unevenly, Negative giving up 0.042 F1 and Positive 0.026 while Neutral moves only 0.003. That the Irrelevant category is hardest to recover is interpretable rather than surprising, since it is a relevance judgement rather than a sentiment.

Fig. 5 shades the same counts, and the shading shows something the digits alone do not: the errors gather into two rows rather than scattering around the diagonal. The Negative

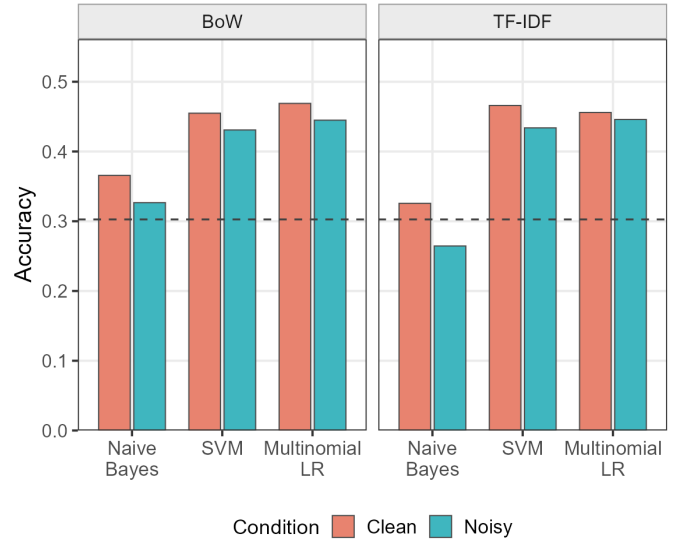


Fig. 4. Clean and noisy accuracy for all six representation–classifier pairs, from Table III. The dashed line is the 30.26% majority-class baseline. Between the panels only the Naive Bayes group changes appreciably, whereas within each panel Naive Bayes is separated from the other two classifiers under both conditions.

TABLE IV
CONFUSION MATRIX FOR MULTINOMIAL LR + BOW ON THE CLEAN TEST SET ($n = 998$); ROWS ARE PREDICTIONS, COLUMNS ARE TRUE LABELS

Predicted	True label				
	Irrel.	Neg.	Neut.	Pos.	Total
Irrelevant	48	18	18	25	109
Negative	47	164	60	48	319
Neutral	26	44	100	46	216
Positive	53	76	69	156	354
Total	174	302	247	275	998

TABLE V
PER-CLASS PRECISION, RECALL AND F1 FOR MULTINOMIAL LR + BOW, DERIVED FROM (2)

Cond.	Class	Support	Prec.	Rec.	F1
Clean	Irrelevant	174	0.440	0.276	0.339
	Negative	302	0.514	0.543	0.528
	Neutral	247	0.463	0.405	0.432
	Positive	275	0.441	0.567	0.496
Noisy	Irrelevant	174	0.429	0.259	0.323
	Negative	302	0.476	0.497	0.486
	Neutral	247	0.461	0.401	0.429
	Positive	275	0.413	0.546	0.470

and Positive prediction rows take 673 of the 998 tweets between them, each drawing at least 47 cases from every true class, while the Irrelevant row stays pale across its full width. The model has learned two broad attractors and two narrow ones.

		True label			
		Irrelevant	Negative	Neutral	Positive
Predicted label	Irrelevant	48	18	18	25
	Negative	47	164	60	48
	Neutral	26	44	100	46
	Positive	53	76	69	156

Fig. 5. The confusion matrix of Table IV with cells shaded by count; diagonal entries are set in bold. The shading exposes the Negative and Positive prediction rows as the two sinks that absorb most of the misclassified tweets.

E. Stability Under Cross-Validation

Table VI reports every individual fold of the 5-fold cross-validation on clean data. Reporting folds separately rather than only their mean exposes the fold-to-fold spread and shows that one Naive Bayes fold (TF-IDF, fold 4, 29.60%) falls below the majority-class baseline entirely. Two independent stability checks support the ordering: the repeated noise trials give a standard deviation of at most 1.1 percentage points at any severity, and the cross-validation standard deviations range from 0.78 to 1.83 points. The 10-to-18-point classifier gap is an order of magnitude larger than either figure, whereas the 1.4-point representation gap is of the same order as the fold-to-fold noise, which is precisely why the first ordering holds and the second does not. A paired t -test over the five folds was deliberately avoided: as Dietterich [23] showed, folds share most of their training data, so such a test violates the independence assumption and has an elevated Type I error rate.

Fig. 6 draws the same folds as bars, which makes the two kinds of variation directly comparable. The Naive Bayes bar is the shortest of its group in every fold of both panels, whereas SVM and MLR are separated by 2.4 points at most and are tied to two decimal places in two BoW folds. Each model also moves by only 1.85 to 4.20 points across its own five folds, so a different partition of the data could not plausibly close the Naive Bayes gap.

F. Comparison with Related Work

The ordering obtained here agrees with Srikanth et al. [8], who also found Logistic Regression strongest among Naive Bayes, a linear SVM, and Logistic Regression on TF-IDF features, and with Al Sarayrah et al. [9], whose Naive Bayes variants trailed their RBF-kernel SVM by 7.2 and 8.4 points. The benchmark of Reusens et al. [2] reaches the same conclusion about TF-IDF pipelines at much larger scale. The degradation observed for Naive Bayes is consistent

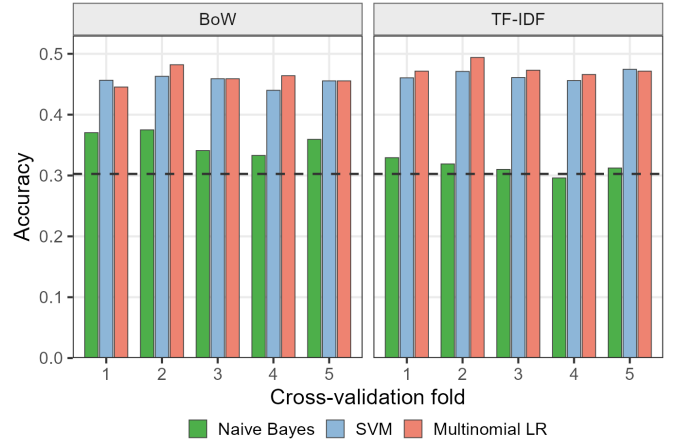


Fig. 6. Per-fold cross-validation accuracy from Table VI, with the 30.26% majority-class baseline dashed. Naive Bayes is the shortest bar of its group in all ten fold-by-representation cells; SVM and MLR are never more than 2.4 points apart and swap order between folds.

with Belinkov and Bisk [11], Moradi and Samwald [12], and Aliakbarzadeh et al. [13], who report that models depending on exact token matching are disproportionately sensitive to small input perturbations, although those studies concern neural models. Internally, repeating the noise injection five times per severity and cross-validating the clean-data ranking is what separated the unstable representation comparison from the stable classifier comparison; a single-run study would have reported the 1.3-point BoW advantage for MLR as a result.

V. CONCLUSION AND FUTURE WORK

This study compared BoW and TF-IDF, paired with Naive Bayes, a linear SVM, and Multinomial Logistic Regression, under controlled textual noise on a four-class Twitter sentiment task. Accuracy fell steadily across 4,998 stratified tweets as the word-deletion probability rose from 10% to 30%, but the clearest practical message is one of priority. For the two discriminative classifiers the representation moved accuracy by at most 1.4 percentage points and reversed direction between the single split and cross-validation, whereas the classifier moved it by 10 to 18 points in a direction that held in every fold. Naive Bayes was the only pairing in which the representation carried real weight, and it was already the weakest of the three under both; on noisy TF-IDF it fell to near-chance agreement ($\kappa = 0.046$). For a lightweight, non-neural sentiment pipeline on noisy social-media text, MLR or an SVM should therefore be preferred, and either weighting scheme is then acceptable.

Three limitations apply: the 177-term vocabulary left by sparsity filtering bounds absolute accuracy; that filtering and the document frequencies in (1) used the full corpus, giving all six pairs a small optimistic bias; and Naive Bayes was fitted with Gaussian rather than multinomial densities [16], so part of its weakness is a model-family mismatch.

Future work will refit the vocabulary and the inverse document frequencies on the training split alone under a relaxed

TABLE VI
PER-FOLD 5-FOLD CROSS-VALIDATION ACCURACY ON CLEAN DATA (%). BASELINE ACCURACY IS 30.26%

Representation	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
BoW	Naive Bayes	37.04	37.50	34.10	33.30	35.94	35.57	1.83
BoW	SVM	45.65	46.30	45.90	44.00	45.55	45.48	0.88
BoW	Multinomial LR	44.54	48.20	45.90	46.40	45.55	46.12	1.35
TF-IDF	Naive Bayes	32.93	31.90	31.00	29.60	31.23	31.33	1.23
TF-IDF	SVM	46.05	47.10	46.10	45.60	47.45	46.46	0.78
TF-IDF	Multinomial LR	47.15	49.40	47.30	46.60	47.15	47.52	1.08

sparsity threshold, to test whether the 46.9% ceiling is an artefact of this feature space, and will replace Gaussian with multinomial Naive Bayes. The two noise mechanisms can then be separated to identify the more damaging one.

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